Asignación #2

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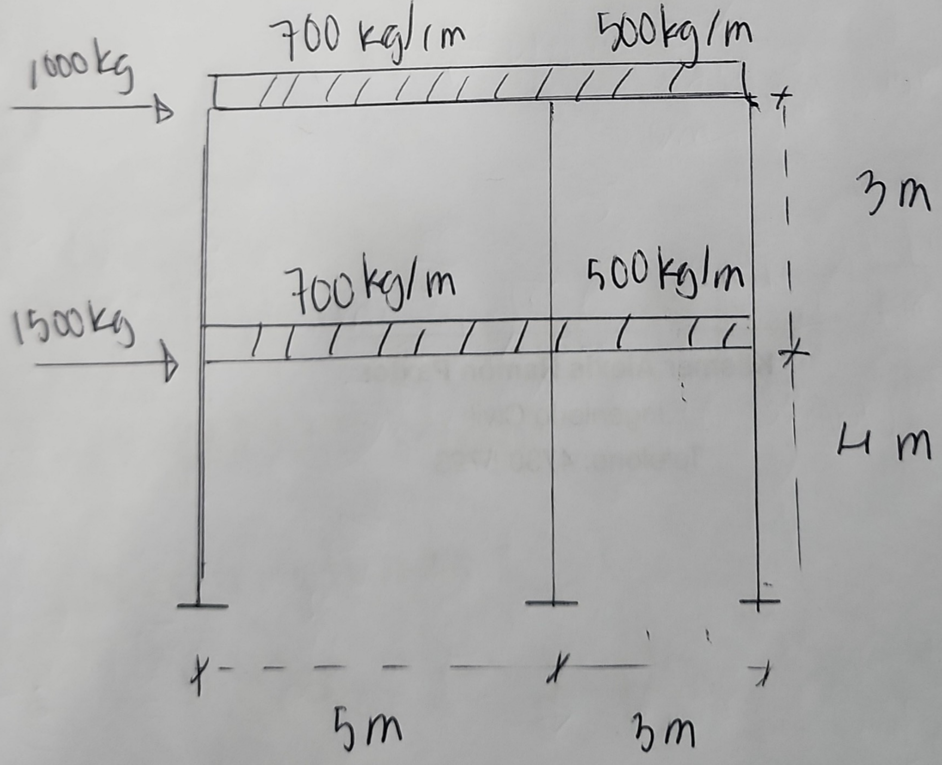
Para el marco que se muestra en la figura, usando el método de la rigidez viga columna bidimensional, encontrar

- Desplazamiento de cada grado de libertad
- Reacción en los apoyos.
- Fuerzas internas en cada elemento.



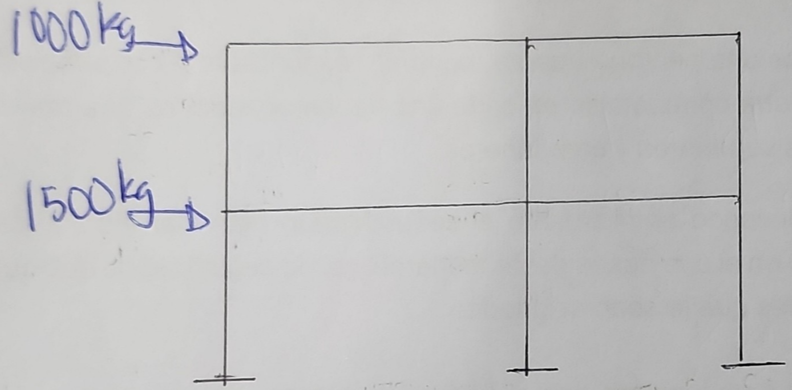
Datos:

columnos: 25 X 25 cm

Vigos : 25 x 50 cm

Concreto: 210 kg/cm²

Fuerzos Externos Previas



Propiedad de los Elementos

Modulo de Elasticidad:

$15,100 \sqrt{f'c} \rightarrow 15,100 \sqrt{210 \text{ kg/cm}^2}$

$= 218819.78 \text{ kg/cm}^2$

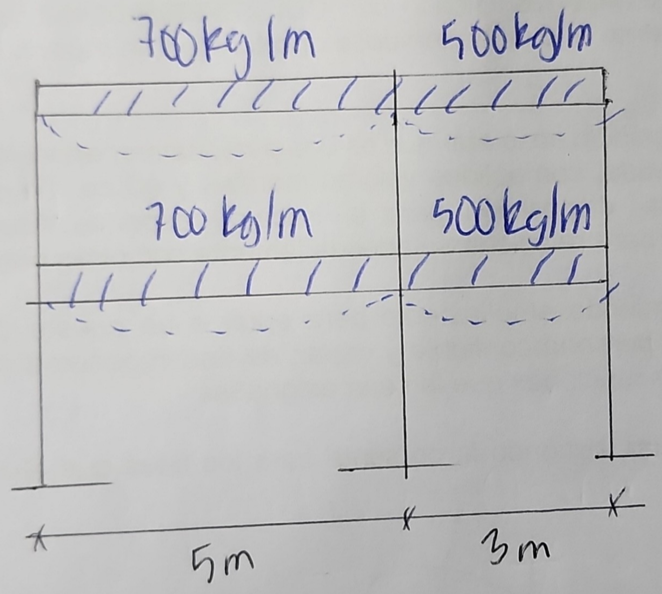
Inercia Columna:

$Ic = \frac{b \cdot h^3}{12} \rightarrow Ic = \frac{25(25)^3}{12} \rightarrow Ic = 32552.08 \text{ cm}^4$

Inercia Viga:

$$I_v = \frac{b \cdot h^3}{12} \rightarrow I_v = \frac{25(50)^3}{12} \rightarrow I_v = 260416.67\ cm^4$$

Convertir Carga
Distribuido → Puntual



Carga Puntual aplicada al centro del tramo

$\boxed{w = 700\ kg/m}$

$$P = \frac{w \cdot L}{2}$$

$$P = \frac{700\ kg/m * 5m}{2}$$

$P = 17500.kg$ ✓

$\boxed{w = 500\ kg/m}$

$$P = \frac{w \cdot L}{2}$$

$$P = \frac{500\ kg/m * 3m}{2}$$

$P = 750kg$ ✓

Carga

Momentos

$\boxed{W = 700 kg/m}$

$$M = \frac{WL^2}{12}$$

$$M = \frac{700kg/m \times 5m^2}{12}$$

$$M = 1458.33 kg \cdot m$$

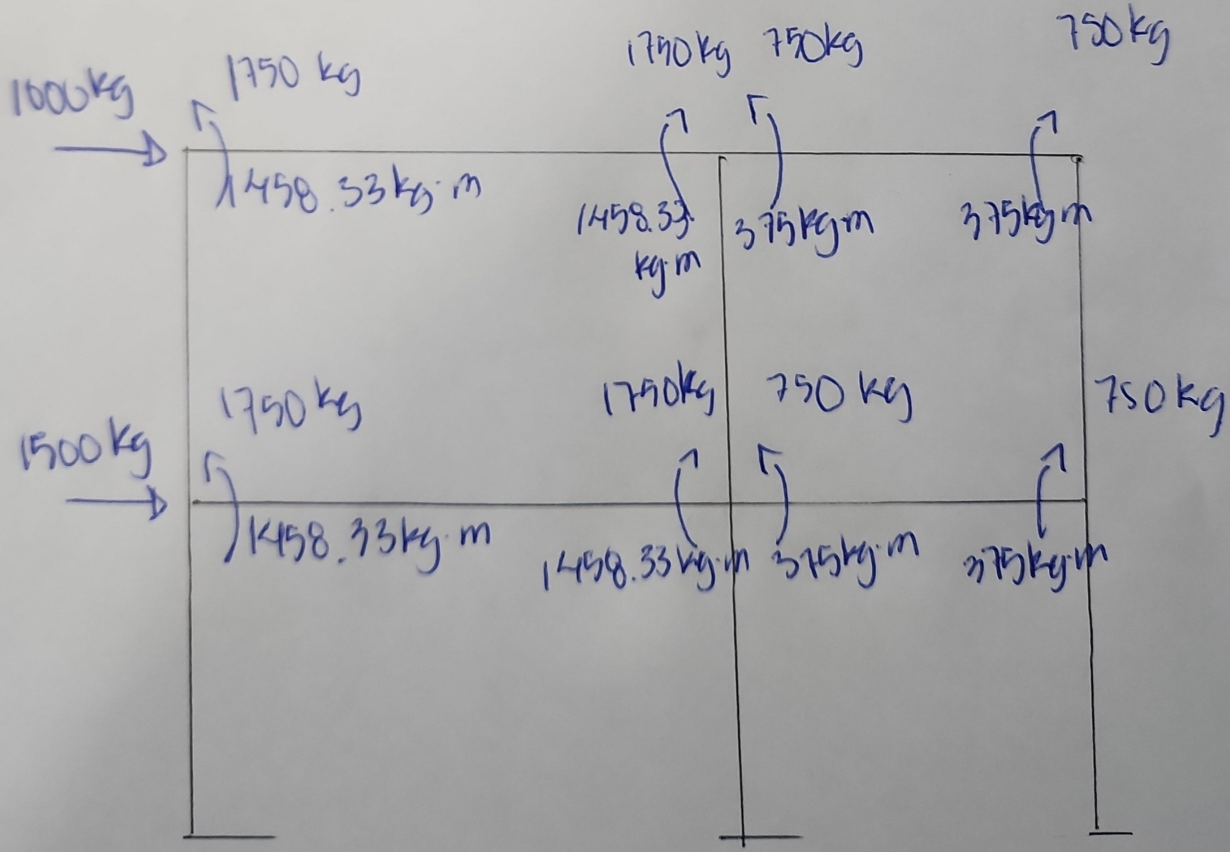
$W = 500 kg/m$

$$M = \frac{ML^2}{12}$$

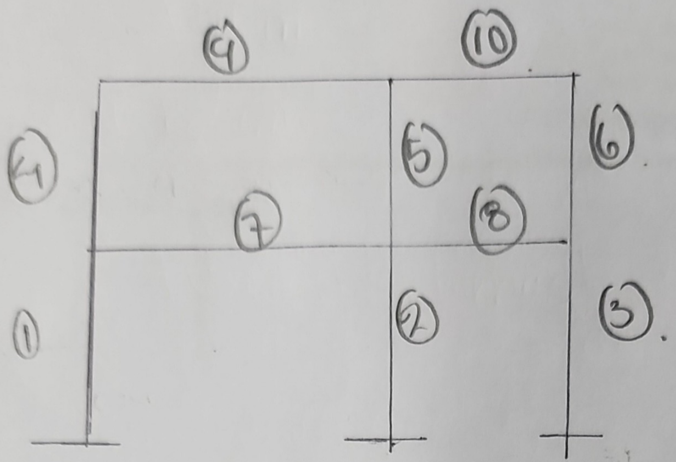
$$M = \frac{500kg/m^2 \times 3m^2}{12}$$

$$M = 375 kg \cdot m.$$

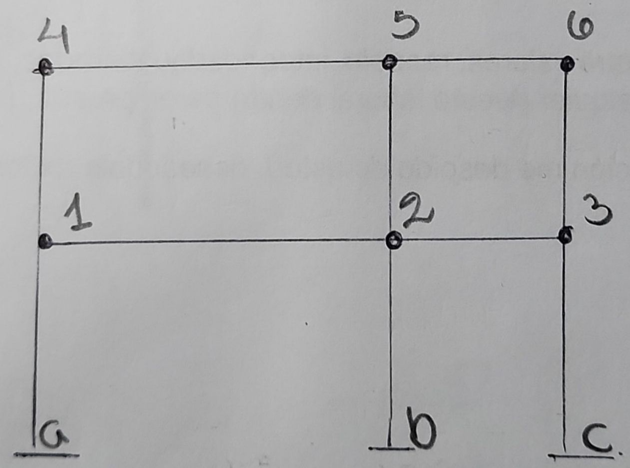
Fuerzas Externas Finales

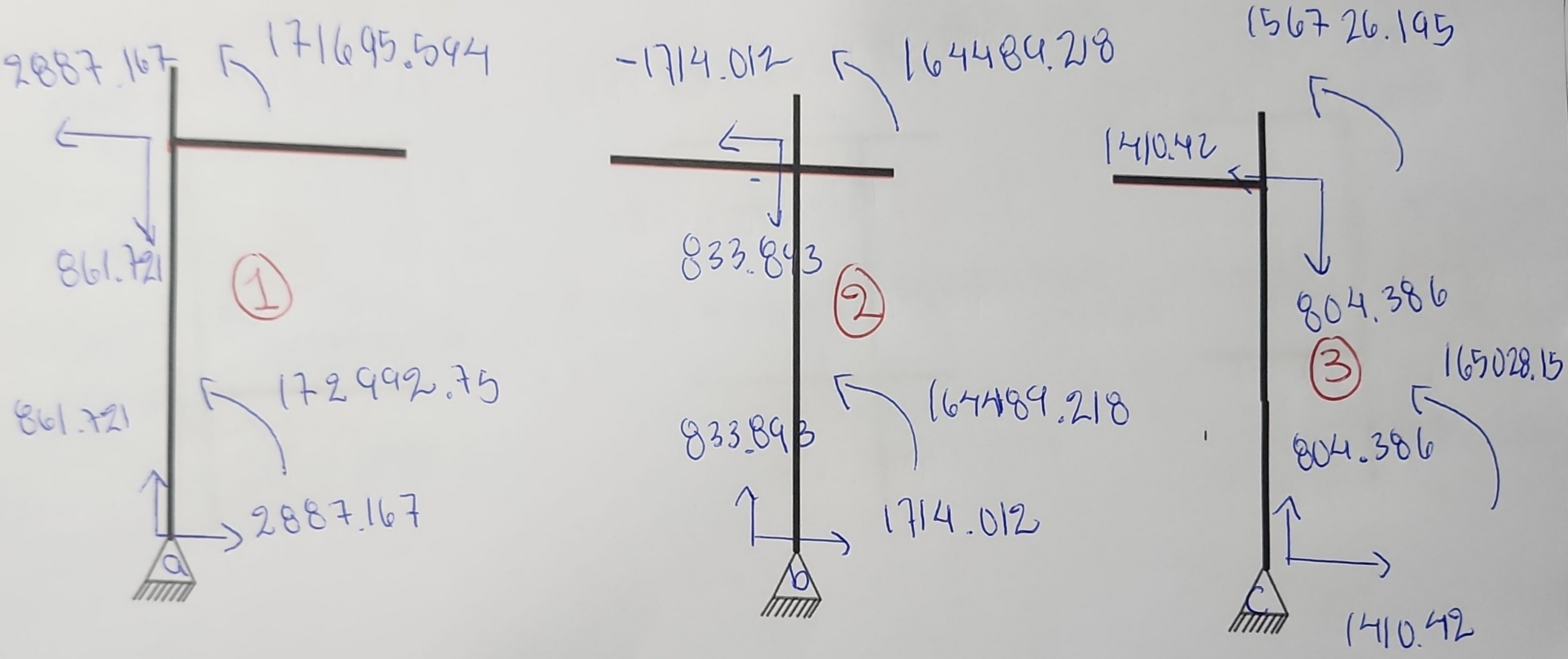


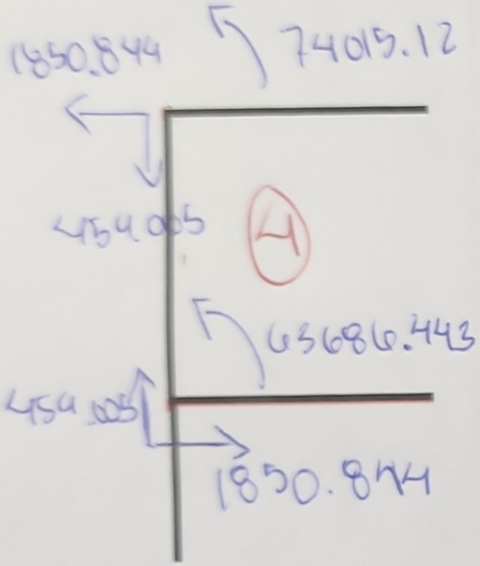
Enumerar los elementos

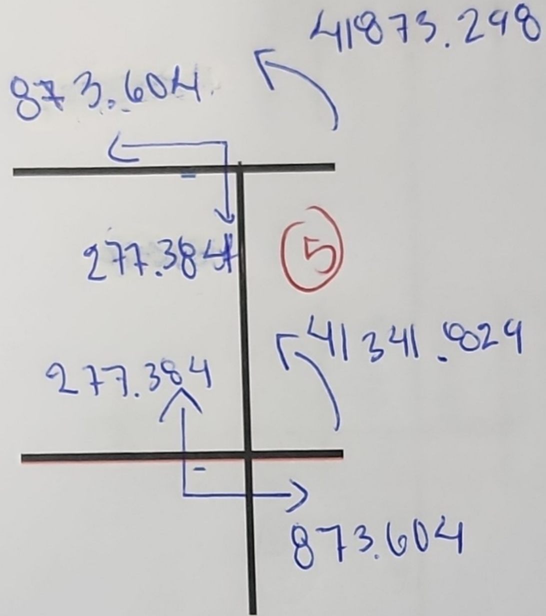


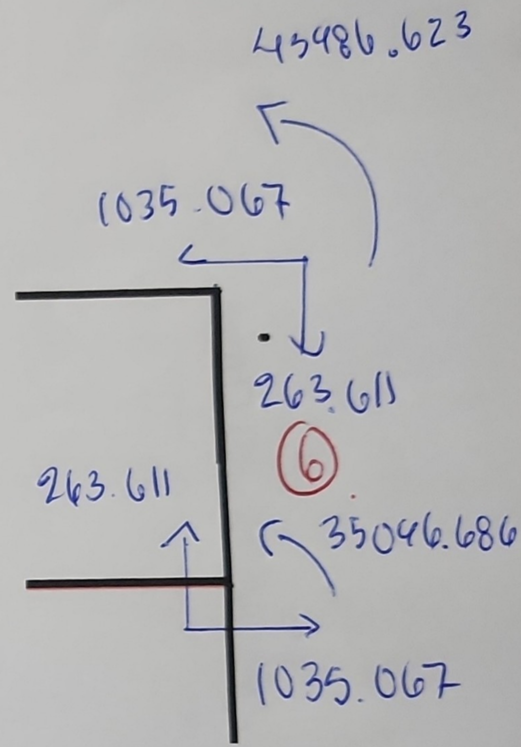
Enumerar grados de libertad



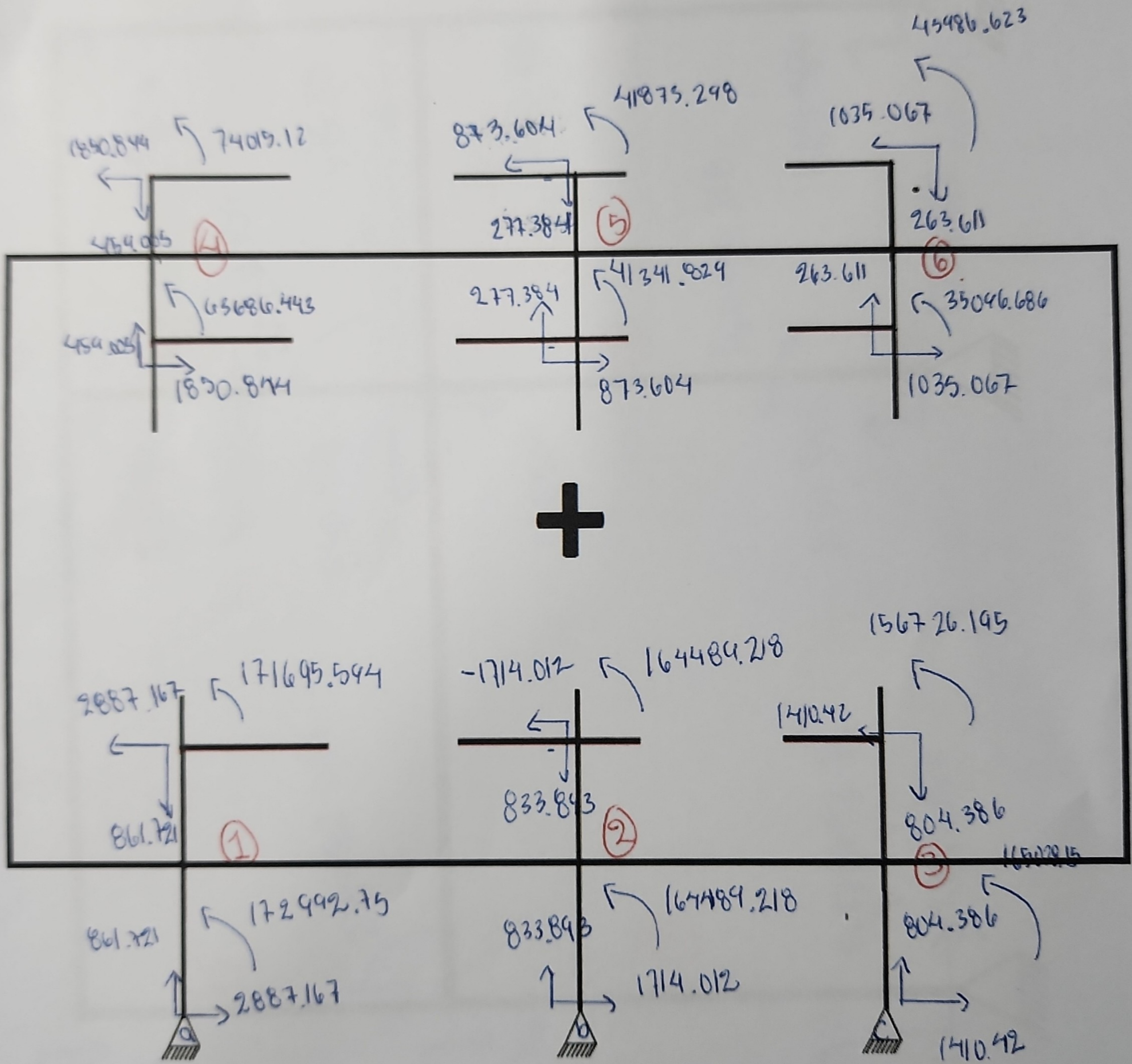

2887.167
171695.544
861.721
1
861.721
172992.75
2887.167
a
-1714.012
164484.218
833.843
2
833.843
164484.218
1714.012
b
156726.195
1410.42
804.386
3
165028.15
804.386
1410.42
c


1850.844
74015.12
454.005
4
63686.443
454.005
1850.844

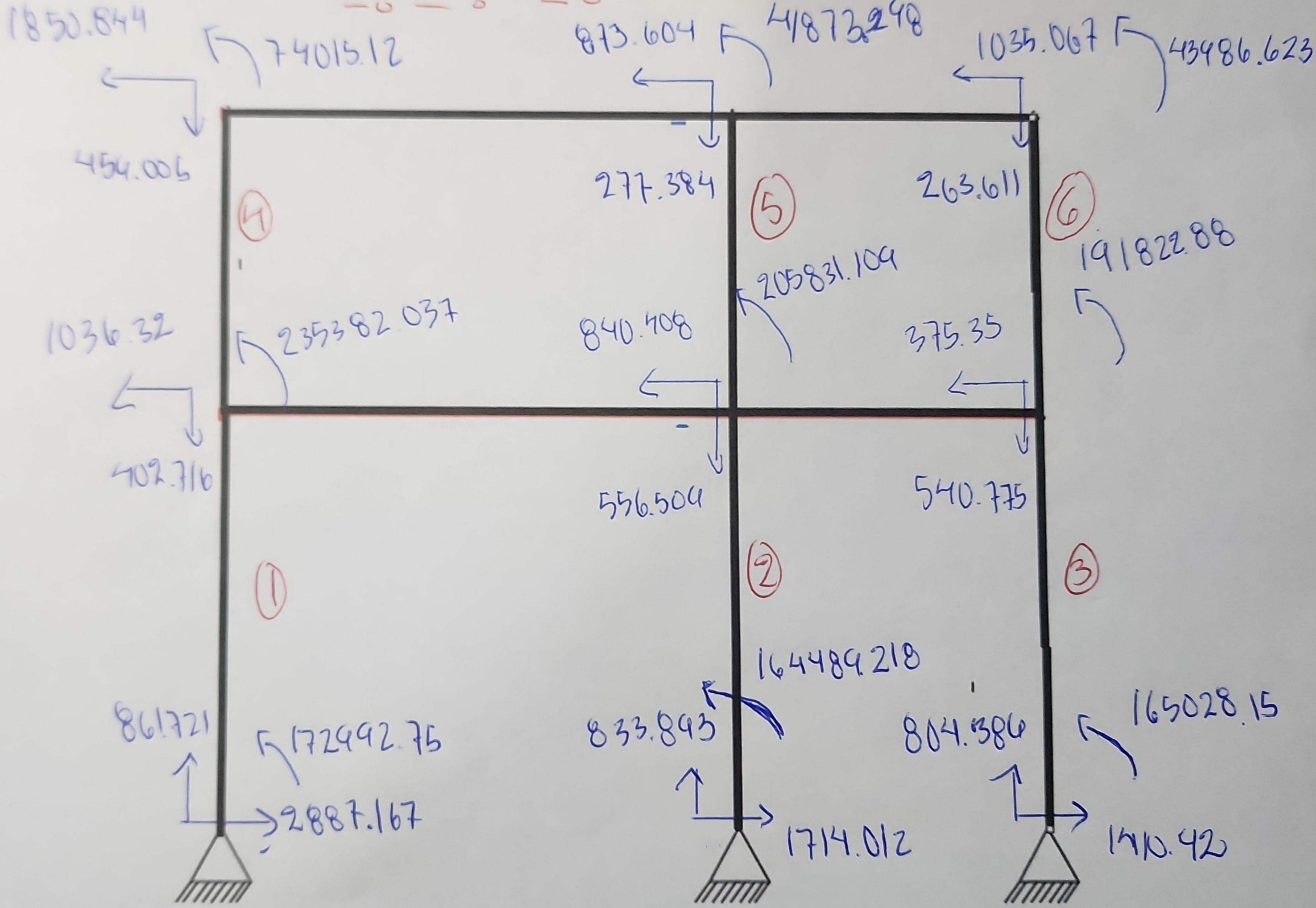

873.604
41873.248
277.384
5
41341.824
277.384
873.604


45486.623
1035.067
263.611
6
263.611
35046.686
1035.067

Se sumaron elementos intermedios



Resultados



1850.844
74015.12
454.006
873.604
41873.248
277.384
1035.067
43486.623
263.611
4
5
6
191822.88
205831.104
1036.32
235382.037
840.408
375.35
402.716
556.504
540.775
1
2
3
164489.218
165028.15
861.721
172492.75
833.843
804.386
2887.167
1714.012
1410.42